

When Does Density Beat Centroids?

A Comparative Study of Clustering Methods Across Data Geometries

[DECIDE: final title]

SC4020 Project 1 — Group XX

[All: full names exactly as on student cards; replace XX with group ID]

Abstract

Owner: Ky (write last)

[Ky: 150–250 words: task, methods compared, datasets, 3 headline findings with numbers, one-line conclusion.]

[DECIDE: Thesis. Proposed: the geometry of the data (cluster shape, density variation, dimensionality, gaps between groups) decides which method works, and internal scores such as silhouette cannot tell you which. Every member’s data supports this (Section 3.4.5).]

1 Introduction

Owner: Wei Yew (draft); Ky to review

1.1 Motivation

Clustering is used where no answer key exists. A seismologist grouping earthquake epicentres wants to find active fault systems. A housing analyst grouping resale transactions wants market segments. Neither has labels to check a result against. Any algorithm they run will still return a partition, along with a score.

The methods differ in what they take a cluster to be. K-means [12] looks for compact groups around centroids and assigns every point to one of them. DBSCAN [8] looks for dense regions separated by sparse space and leaves isolated points as noise. Gaussian mixtures stretch the K-means model to ellipses. Spectral clustering drops the shape assumption and works on a similarity graph. When a method’s assumption matches the data, it recovers real structure. When it does not, it still returns clusters. On one year of global earthquakes, K-means places five plate-boundary belts, lying 2,932 to 11,449 km apart, in a single cluster (Section 3.4.1).

The practical question is which method fits which data. It has to be answered without labels.

1.2 Problem statement

Without labels, methods and parameters are usually chosen by an internal index. The most common is the silhouette coefficient [16]. It compares each point’s mean distance to its own cluster with its mean distance to the nearest other cluster. It is highest for compact, well-separated groups: the cluster model of K-means. Choosing between K-means and a density method by silhouette is therefore not a neutral test.

The bias is large on our data. On Wine, DBSCAN scores a higher silhouette than K-Means++ (0.42 against 0.28) while agreeing far less with the true classes (ARI 0.31 against 0.90). It earns that score by labelling 65% of the wines as noise, which the silhouette leaves out. On Ecoli, the silhouette ranks four methods in almost the reverse of their ARI order (Spearman $\rho = -0.80$). On HDB resale flats, K-means on unscaled features scores 0.540 by splitting sales into four price bands, because price alone sets the distances.

The data can also mislead before any method runs. Treated as flat coordinates, latitude and longitude put 179.9°E and 179.9°W at opposite edges of the map, although they lie about 20 km apart. Storey numbers recorded in three-floor bands create thousands of identical values, which density methods read as dense clusters.

This report asks two questions. Which properties of a dataset decide whether a centroid, mixture, density or graph-based method succeeds? What can an analyst check without labels to avoid a misleading choice?

1.3 Contributions

[All: update dataset and method counts once Table 2 is final.]

- A comparison of four core methods (K-Means, K-Means++, DBSCAN, HDBSCAN) on eight datasets, extended with GMM, spectral and agglomerative clustering. Results are grouped by the data property that decides the outcome: cluster shape, density variation, dimensionality and the presence of gaps between groups (Section 3.4).
- Evidence from five datasets that the silhouette can prefer the wrong answer (Section 3.4.5), with a fake-data check that catches data containing no structure to find (Section 3.3). On HDB, only K-means beats data whose columns are shuffled independently (silhouette 0.317 against 0.218). Both density methods score below their fake-data baseline.
- Ablations that change one component at a time: initialisation, coordinate handling, feature scaling, dimensionality and values recorded in steps (Section 3.5). On Ecoli, DBSCAN and K-Means++ swap rank between three and four PCA dimensions.
- Seven implemented improvements, each tied to an observed failure and a measured effect (Section 3.6). Surface distance keeps together all 92 close earthquake pairs that straddle the 180° meridian; flat coordinates keep none. HDBSCAN keeps 62% of the sparse belts in the Americas, against 20% for DBSCAN with one global radius.
- Runtime on all 234,294 HDB sales (Section 3.7). K-means finishes in under 0.06 s and HDBSCAN in 212 s; DBSCAN would need about 6 GB for its neighbour lists. Section 3.9 turns these results into selection guidance.

1.4 Related work

Centroid and mixture models. Lloyd’s algorithm [12] minimises the within-cluster sum of squared distances by alternating assignment and centroid updates. It converges to a local optimum that depends on the starting centres. K-means++ [2] picks each starting centre with probability proportional to its squared distance from the centres already chosen, which gives an $O(\log k)$ approximation to the optimum in expectation. A Gaussian mixture fitted by EM [7] generalises K-means to ellipsoidal clusters with soft assignments. Agglomerative clustering builds a hierarchy by repeatedly merging the closest pair of clusters. Under Ward’s criterion [18] the merge cost is the increase in within-cluster variance, the same objective as K-means.

Density-based methods. DBSCAN [8] defines a cluster as a maximal set of density-connected points. A point is a core point if at least `minPts` points lie within radius ϵ . Points reachable from no core point are noise. The method needs no k and imposes no shape, but its single global ϵ assumes all clusters have similar density. OPTICS [1] orders points by reachability so that clusters at several densities can be read from one plot. HDBSCAN [5] builds a hierarchy over mutual reachability distances and keeps the clusters that persist longest. This removes ϵ and leaves a minimum cluster size as the main parameter.

Graph-based methods. Spectral clustering [14] embeds the points using the leading eigenvectors of a normalised affinity matrix, then runs K-means in that embedding. Groups connected through the affinity graph become compact in the embedding, so non-convex clusters can be separated. A dense affinity matrix needs $O(n^2)$ memory, which limits the method to modest n .

Validation and cluster tendency. Internal indices, such as the silhouette [16], Davies–Bouldin [6] and Calinski–Harabasz [4] indices, score a partition from the data alone. External indices such as the Adjusted Rand Index [11] compare it with known labels. Von Luxburg et al. [13] argue that no criterion can judge a clustering apart from its intended use. A separate line of work tests whether data contain clusters at all. The Hopkins statistic [9] compares nearest-neighbour distances in the data with those in uniformly random points. The gap statistic [17] compares within-cluster dispersion with its value on reference data drawn from a null distribution. Our fake-data check applies the same idea to every method we compare. In high dimensions the ratio of nearest to farthest neighbour distance tends to one [3], which weakens any method built on a density threshold.

2 Methods

Owner: Ky (K-Means, K-Means++, DBSCAN, HDBSCAN); Lam (GMM, Spectral); Hoang Anh (Agglomerative)

2.1 K-Means and K-Means++

[Ky: Port Sec. 3.1.]

2.2 DBSCAN

[Ky: Port Sec. 3.2.]

2.3 HDBSCAN

[Ky: Port Sec. 3.3.]

2.4 Gaussian Mixture Model

[Lam: EM fit, covariance type used, parameters, complexity, assumptions.]

2.5 Spectral clustering

[Lam: Affinity graph, eigenvectors, K-means step; affinity used per dataset.]

2.6 Agglomerative clustering

[Hoang Anh: Linkage used and why; parameters; complexity.]

3 Experiments

3.1 Datasets

Owner: each dataset’s owner writes its paragraph (max. 120 words)

Table 1: Structural assumptions of the methods compared.

	K-Means	K-Means++	DBSCAN	HDBSCAN	GMM	Spectral	Agglom.
Cluster shape assumed	isotropic	isotropic	arbitrary	arbitrary	[Lam: ?]	[Lam: ?]	[HA: ?]
Requires k	yes	yes	no	no	[Lam: ?]	[Lam: ?]	[HA: ?]
Explicit noise label	no	no	yes	yes	[Lam: ?]	[Lam: ?]	[HA: ?]
Handles varying density	no	no	no	yes	[Lam: ?]	[Lam: ?]	[HA: ?]
Parameter is interpretable	no	no	yes (km)	partly	[Lam: ?]	[Lam: ?]	[HA: ?]
Time complexity	$O(nkdi)$	$O(nkdi)$	$O(n \log n)^a$	$O(n \log n)^a$	[Lam: ?]	[Lam: ?]	[HA: ?]

^a With an effective spatial index; $O(n^2)$ worst case.

Table 2: Datasets, grouped by the data property each one tests. [DECIDE: final list; drop or keep Mall; Ky’s 1965–2016 earthquakes replaced by the 2023 surface-distance pipeline?]

ID	Dataset	Owner	n	d	Labels	Property tested
D1	make_moons (synthetic)	Lam	1,000	2	2 classes	curved shapes (control)
D2	Earthquakes, 2023, $M \geq 4.5$	Wei Yew	7,643	2	none	long belts, uneven density
D3	Uber NYC pickups, Apr 2014	Ky, Hoang Anh	[DECIDE: 6k or 15k]	2	none	density contrast in a city
D4	Ecoli	Ky	336	7	8 classes	dimensionality, rare classes
D5	Fashion-MNIST (PCA)	Lam	3,000	50	10 classes	high dimensions
D6	Wine	Lam, Hoang Anh	178	13	3 classes	silhouette vs labels
D7	Breast cancer (Wisconsin)	Hoang Anh	569	30	2 classes	silhouette vs labels
D8	HDB resale flats, 2017–2026	Wei Yew	20,000 ^b	4	flat type (7)	no gaps, scale, recorded ties

^b Random sample of 234,294 usable sales; all 234,294 used for runtime (Section 3.7).

D1 make_moons. [Lam: generator settings (noise 0.07), why it is the shape control.]

D2 Earthquakes. [Wei Yew: USGS ComCat 2023, $M \geq 4.5$; 8 volcanic events removed; depth not clustered (44% default 10 km). Source: [weiyew/sc4020-earthquake/notebooks/eda.ipynb](https://github.com/weiyew/sc4020-earthquake/notebooks/eda.ipynb).]

D3 Uber NYC. [Ky + Hoang Anh: one merged paragraph; agree sample size and bounding box.]

D4 Ecoli. [Ky: Port from Sec. 4.1.]

D5 Fashion-MNIST. [Lam: 3,000-image sample, PCA to 50 dims; why subsampled.]

D6 Wine. [Lam + Hoang Anh: one merged paragraph; both runs give K-Means ARI 0.897.]

D7 Breast cancer. [Hoang Anh: 30 features, standardised; benign vs malignant.]

D8 HDB resale. [Wei Yew: 4 features (floor area, remaining lease, storey, index-adjusted price), standardised; flat type held out as label.]

3.2 Preprocessing

Owner: Wei Yew

[Wei Yew: Three rules applied to every dataset where relevant: (i) coordinates on the sphere (haversine for density methods, 3D unit vectors for K-means); (ii) standardise features; (iii) check for values recorded in steps and jitter within the step. Each rule is tested in Section 3.5.]

3.3 Evaluation metrics and parameter selection

Owner: Ky, Wei Yew

Metrics. [**Ky + Wei Yew:** Internal: silhouette, Davies–Bouldin, Calinski–Harabasz (noise excluded). Coverage: noise %, largest-cluster %. External (labelled data): ARI, NMI (noise kept as a label). Fake-data check: same method on shuffled columns (tabular) or uniform points (coordinates); a result counts only if the real data scores clearly higher. Runtime.]

Selection rule. [**DECIDE:** One rule for every dataset and method. Ky: ≥ 3 clusters, $\leq 40\%$ noise. Wei Yew: ≥ 2 clusters, $\leq 50\%$ noise. Add a largest-cluster cap? (It picks storey-band artefacts on HDB, Section 3.5.)] [**All:** Labelled datasets: report both the rule’s choice and the best-ARI (oracle) choice, so the gap between them is visible.]

Parameter grids. [**All:** One line per method: k range; ε grid (k-distance knee or quantiles); minPts; min_cluster_size and min_samples; GMM components; spectral affinity.]

Reproducibility. [**DECIDE:** shared code: Ky’s src/core.py or Wei Yew’s metrics.py, so every table has the same columns.] [**All:** scikit-learn version, seeds, hardware.]

3.4 Comparison by data property

Owner: All

[**All:** One results table per property (rows: dataset $\times$ method; columns: #clusters, noise, S, DB, ARI, NMI, time, fake-data gap). Bold the best per dataset.]

3.4.1 Curved and elongated clusters

Owner: Lam (D1), Wei Yew (D2)

[**Lam:** D1: DBSCAN ARI 0.982, Spectral 1.000, K-Means++ 0.481 (lam/results/make_moons_results.csv).] [**Wei Yew:** D2: K-means puts 5 belts 2,932–11,449 km apart into one cluster.]

3.4.2 Uneven density

Owner: Wei Yew (D2), Ky and Hoang Anh (D3)

[**Wei Yew:** D2: one ε keeps 90% of the dense south-west Pacific but 20% of the sparse Americas; HDBSCAN keeps 62% of the Americas.] [**Ky + Hoang Anh:** D3: Midtown vs outer boroughs; knee ε vs silhouette-optimal ε .]

3.4.3 High dimensions

Owner: Ky (D4), Lam (D5)

[**Ky:** D4: crossover between 3 and 4 PCA dimensions (Ablation D).] [**Lam:** D5: DBSCAN ARI 0.033; HDBSCAN labels 71% of points noise.]

3.4.4 No gaps between groups

Owner: Wei Yew (D8)

[**Wei Yew:** D8: only K-means beats the shuffled null (silhouette 0.317 vs 0.218); DBSCAN puts 97.1% of sales in one cluster; both density methods score below their null.]

3.4.5 Silhouette versus ground truth

Owner: All

[All: One paragraph, one line per dataset: D4 best silhouette (HDBSCAN) has the worst ARI, rank correlation -0.80 (Ky); D6 DBSCAN silhouette 0.42 vs K-Means++ 0.28 , ARI 0.31 vs 0.90 (Lam); D7 K-Means top silhouette 0.343 , GMM best ARI 0.774 vs 0.654 (Hoang Anh); D8 unscaled K-means silhouette 0.540 from price bands (Wei Yew); D5 HDBSCAN top silhouette with 71% noise (Lam).]

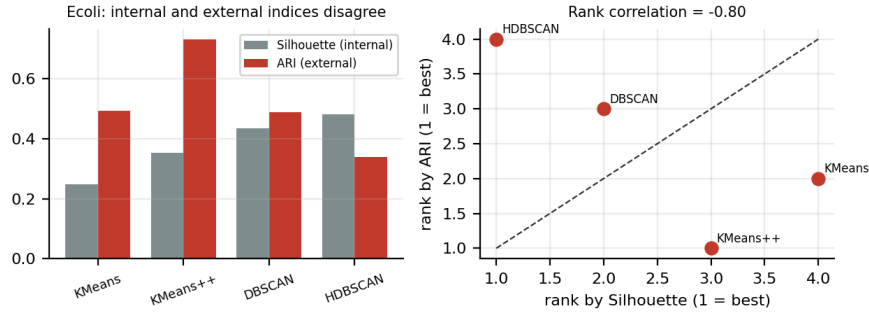


Figure 1: [Ky: existing figure; extend to D5–D8 or make one panel per dataset.]

3.5 Ablation studies

Owner: Ky, Wei Yew

3.5.1 A: Initialisation (K-Means vs K-Means++)

[Ky: Port Ablation A (30 seeds, D3/D4). Add ARI mean $\pm$ s.d. over the 30 seeds: the 0.24 ARI claim currently rests on seed 42 only.] [Wei Yew: D2: worst single-start run 18.8% (random) vs 11.5% (K-Means++) above the best. D8: no difference, all 40 runs within 0.009% of the best.]

3.5.2 B: Distance on geographic coordinates

[Wei Yew: Flat degrees vs sphere on D2: 92 close pairs straddle the 180° line; sphere keeps 92 (K-means), 86 (DBSCAN), 79 (HDBSCAN) together, flat degrees 0. K-means silhouette $0.528 \rightarrow 0.404$. Source: [weiyew/sc4020-earthquake/notebooks/ablation.ipynb](https://github.com/weiyew/sc4020-earthquake/notebooks/ablation.ipynb), Sec. 2.] [DECIDE: Ky’s projected-km pipeline has the same $\pm 180^\circ$ cut. Rerun D3 on the sphere, or state that city scale makes it irrelevant.]

3.5.3 C: Feature scaling

[Ky: D4: standardisation lifts K-Means++ ARI $0.497 \rightarrow 0.731$; DBSCAN unaffected once ε is re-tuned.] [Wei Yew: D8: raw units give price bands (silhouette 0.540 , ARI 0.261 vs standardised).]

3.5.4 D: Dimensionality

[Ky: Port Ablation D. Call it PCA dimension; “intrinsic dimensionality” overclaims.]

3.5.5 E: Values recorded in steps

[Wei Yew: D8 storey bands: with a 50% largest-cluster cap, 27 of 27 DBSCAN and 6 of 7 HDBSCAN clusters are single storey bands; jittering within the band removes them.]

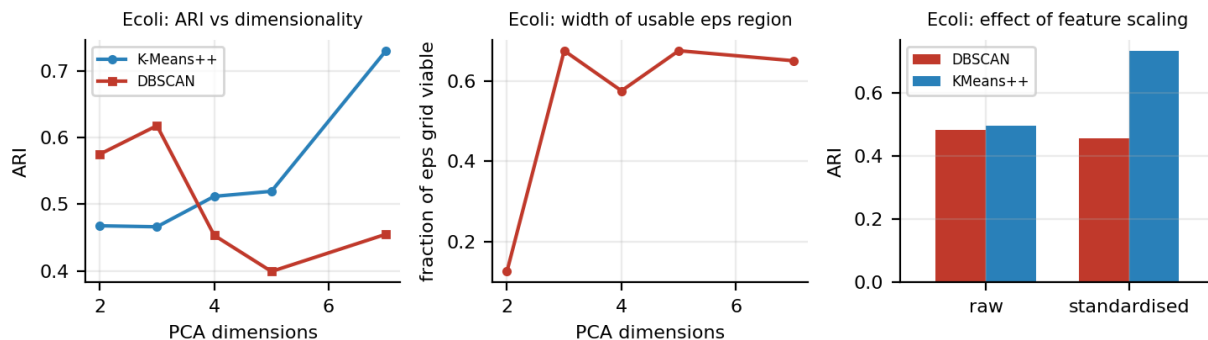


Figure 2: [Ky: existing figure (Ablations C and D on D4).]

3.5.6 F: Data subsets

[Wei Yew: D2: magnitude 5.0+ behaves like a random subset of equal size; half-year vs full-year HDBSCAN ARI 0.41–0.54.]

3.6 Improvements to the methods

Owner: All

Table 3: Implemented improvements. Each fix targets a failure observed in Section 3.4.

Failure	Cause	Fix	Data	Before → after	Owner
Clusters split at 180°	flat lat/lon	surface distance	D2	0 → 92 of 92 pairs kept	Wei Yew
Poor local optimum	random seeding	K-Means++	D4	ARI 0.493 → 0.731	Ky
Sparse belts lost	one global ϵ	HDBSCAN	D2	Americas kept 20% → 62%	Wei Yew
One feature dominates	unequal scales	standardisation	D4	ARI 0.497 → 0.731	Ky
Clusters are storey bands	recorded steps	jitter within step	D8	27 of 27 band clusters → none	Wei Yew
Curved clusters cut	convex assumption	spectral clustering	D1	ARI 0.481 → 1.000	Lam
Degenerate k chosen	silhouette alone	constraints + fake-data check	D2, D3	[Ky: ?]	Ky, Wei Yew

[All: One short paragraph per row: why the fix works, and its cost.] [All: Proposed but not implemented: e.g. ground truth for D2 from plate-boundary maps; OPTICS; approximate spectral clustering for large n .]

3.7 Runtime and scalability

Owner: Wei Yew, Lam

[Wei Yew: D8, all 234,294 sales: K-means under 0.06s; HDBSCAN 212s; DBSCAN not run (neighbour lists need about 6 GB). Plot runtime against n (weiyew/sc4020-hdb/experiments/scalability.py).] [Lam: Why D5 needed subsampling for Spectral, DBSCAN and HDBSCAN; fit times per method.] [Ky: Existing Fig. 9 runtime at $n \leq 23,412$, if kept.]

Wei Yew: runtime vs number of sales, log–log, one line per method

Figure 3: [Wei Yew: caption]

3.8 Success and failure cases

Owner: each owner: one figure per case, with the cause named in the caption

3.8.1 Successes

[**Wei Yew**: HDBSCAN on D2: follows dense and sparse belts.] [**Lam**: DBSCAN / Spectral on D1.] [**Ky**: K-Means++ on D4: compact classes in 7 dimensions.] [**Wei Yew**: HDBSCAN on D8 floor area + lease: 14 clusters, each one flat design. Frame it as a sanity check: a group-by on flat model and floor area gives similar groups.]

3.8.2 Failures

[**Wei Yew**: K-means on D2: one cluster of 5 separate belts.] [**Wei Yew**: Density methods on D8: one mass, or storey bands.] [**Lam**: DBSCAN on D5.] [**Ky**: HDBSCAN on D4: 37% noise removes rare classes.]



Figure 4: [**All**: caption naming the cause of each case]

3.9 Key factors and practical guidance

Owner: Ky

[**Ky**: Port Sec. 7.3 (key factors) and re-rank them with evidence from all datasets: cluster shape, density variation, dimensionality, gaps between groups, scaling, initialisation, recorded steps, parameter choice.] [**Ky**: Port Sec. 7.4 (guidance). Add: compare every result against fake data (Wei Yew); check for values recorded in steps (Wei Yew); report noise % next to every silhouette.]

4 Conclusion

Owner: Ky

[**Ky**: Summary of findings with numbers; answer the title question.]

Limitations

[**All**: One line each: D2 has no ground truth; D8 has no location feature; oracle ARI on labelled data is optimistic; datasets up to 50 dimensions.]

References

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[Ky + Hoang Anh: Uber source.] [Lam + Hoang Anh: Wine, Mall.] [Hoang Anh: Breast cancer.]